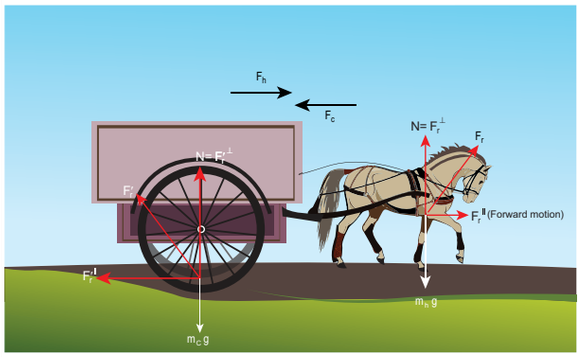
and cart forward with acceleration. As there is an external force acting on the system, Newton's second law has to be applied and not Newton's third law.

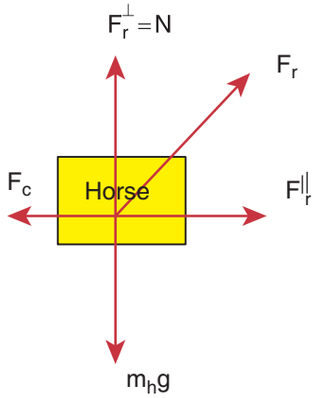
The following figures illustrates this.



If we consider the horse as the 'system', then there are three forces acting on the horse.

(i) Downward gravitational force $(m_h g)$
(ii) Force exerted by the road (F_r)
(iii) Backward force exerted by the cart (F_c)

It is shown in the following figure.



F_r – Force exerted by the road on the horse

F_c – Force exerted by the cart on the horse

$F_r^{\perp}$ – Perpendicular component of $F_r = N$

$F_r^{\parallel}$ – Parallel component of F_r which is reason for forward movement

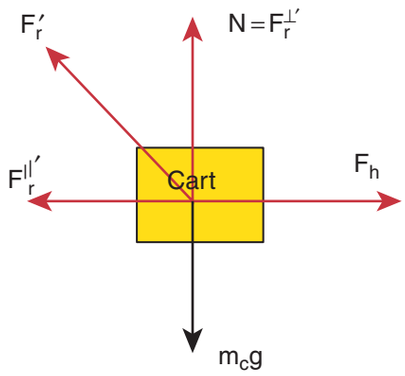
The force exerted by the road can be resolved into parallel and perpendicular components. The perpendicular component balances the downward gravitational force. There is parallel component along the forward direction. It is greater than the backward force (F_c). So there is net force along the forward direction which causes the forward movement of the horse.

If we take the cart as the system, then there are three forces acting on the cart.

(i) Downward gravitational force $(m_c g)$
(ii) Force exerted by the road (F_r')
(iii) Force exerted by the horse (F_h)

It is shown in the figure

Force on the cart



The force exerted by the road $(\vec{F_r'})$ can be resolved into parallel and perpendicular components. The perpendicular component cancels the downward gravity $(m_c g)$. Parallel component acts backwards and the force exerted by the horse $(\vec{F_h})$ acts forward. Force $(\vec{F_h})$ is greater than the parallel component acting in the opposite direction. So there is an overall unbalanced force in the forward direction which causes the cart to accelerate forward.

UNIT-3(XI-Physics_Vol-1).indd 121 28-01-2020 13:47:55